

SM Global Solution Hub — College Project Report

Project type: Full-stack digital marketplace (college / academic demonstration)

Working application: React frontend (`preview/`) + Node.js/Express API (`backend/`) + MongoDB

Local demo: Frontend `http://localhost:5173` · API `http://localhost:5001`

1. Project overview

SM Global Solution Hub is a web platform where:

- Buyers discover ready-made digital projects / products
- Developers list and sell projects on the marketplace
- Clients can request custom work or buy existing products
- Admins manage approvals, leads, quotations, payments, settlements, and reports
- Support / ops staff handle live chat, contact messages, and follow-up reminders

The system is built end-to-end as a **college project / demo product** (not a production commercial launch). Core marketplace, role-based dashboards, payments (test/demo), reports, and live chat are implemented and runnable locally.

2. Problem statement

Small agencies and freelancers often sell digital projects through scattered WhatsApp chats, static websites, and manual Excel tracking. There is no single place for:

- Project catalogue + live demos
- Role-based staff and seller panels
- Lead capture and CRM-style follow-ups
- Quotations → payment → invoice → developer wallet / payout

This project demonstrates a unified hub that covers discovery, sales ops, and support in one application.

3. Objectives

Objective	Status
Public marketplace (browse, detail, contact, customize)	Done
Multi-role authentication (JWT)	Done
Admin / Super Admin operations	Done
Developer listing + wallet / payouts	Done
Client buy / quotations / invoices	Done
Contact inbox + live chat support	Done
CRM follow-ups & reminders	Done
Reports, charts, Excel / PDF exports	Done
Seeded demo data for viva / presentation	Done

4. Tech stack

Layer	Technology
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Frontend	React, Vite, Tailwind CSS, React Router, TanStack Query, Zustand
Backend	Node.js, Express.js
Database	MongoDB Atlas + Mongoose
Auth	JWT with roles
Payments	Razorpay (test) + demo pay path for college demo
Media	Cloudinary (configurable in Super Admin settings)
Exports	ExcelJS, PDFKit

5. User roles in the system

Role	What they can do
Super Admin	Full platform control: users, settings (Razorpay/Cloudinary/GST), payment reports, audit log, all ops modules
Admin	Day-to-day ops: projects, approvals, leads, quotations, invoices, settlements, coupons, reports, live chat, follow-ups
Support Agent	Live chat inbox, leads/requirements, contact inbox, clients/developers directory, follow-ups
Developer	Submit projects, manage requests, wallet & payouts, coupons
Client / Buyer	Requirements, customizations, quotations & pay, invoices; help via Live Chat on the site

6. Major modules / features (what is in this project)

6.1 Public website

- Home / branding pages
- Projects gallery + project detail (including demo links where seeded)
- Developer marketplace / storefront
- Submit requirement & customize project flows
- Contact form → **Contact Inbox** for staff
- Floating **Live Chat** widget (guest must share name, email, phone before chat)

6.2 Authentication & users

- Login / register flows for demo
- Role-protected dashboards
- Staff can create users (as per role rules)
- Users / Clients / Developers directories with detail profiles

6.3 Marketplace & catalogue

- Categories & project CRUD (admin / developer submission)
- Approval workflow for listings
- Curation / reviews (admin modules)

6.4 Sales & money flow

- Quotations (create, send, PDF)
- Client payment (Razorpay test / demo)
- Invoices (PDF)

- Settlements & developer **wallet** (hold → available → payout request)
- Admin approve / mark payout paid
- Coupons
- Super Admin payment reports

6.5 Leads & CRM-style ops

- **Leads / Requirements** list (from requirement & customization forms)
- **Contact Inbox** — one-time contact form messages
- **Follow-ups** — staff-logged call/email/chat notes + next reminder (from client/developer profile)
- **Live Chat** — real-time visitor ↔ team conversation with limited auto-replies; staff can take over

6.6 Reports & analytics

- Ops reports with filters, pagination, Excel export
- Master / platform reports with charts and multi-sheet Excel
- Dashboard KPIs for roles

6.7 Demo / college presentation support

- Seed scripts (`npm run seed` , `npm run seed:money`)
- Demo credentials (see `CREDENTIALS.md`)
- Walkthrough docs under `docs/`

7. System architecture (simple)

```

Browser (React · preview/)
  | REST / JWT
  ▼
Express API (backend/ · port 5001)
  |
  ▼
MongoDB Atlas
  +
Razorpay (test) · Cloudinary · Excel/PDF generation

```

8. Project structure

```

Smm Digital/
├─ backend/          # Express API, models, controllers, seeds
├─ preview/          # Live React UI used for demo
├─ docs/             # API, demo walkthrough, checklists, this report
├─ CREDENTIALS.md    # Demo logins (college only)
├─ PROJECT_PLAN.md   # Original plan
├─ FRONTEND_PLAN.md  # Frontend planning notes
├─ TRACKER.md        # Progress tracker
└─ README.md

```

The production UI for presentation is **preview/** . Early teammate HTML / incomplete frontend drafts were **not** used as the final product.

9. How to run (for viva / lab)

```
# Terminal 1 – API
cd backend
npm install
npm run seed          # optional demo data
npm run dev           # http://localhost:5001

# Terminal 2 – UI
cd preview
npm install
npm run dev           # http://localhost:5173
```

Demo accounts: see `CREDENTIALS.md` .

Suggested 2-minute money demo: `docs/DEMO_WALKTHROUGH.md` .

10. Team contribution & work division

This section is included for college evaluation honesty: **who contributed what**, and what was actually shipped in the final project.

10.1 Team members

Name	Claimed / intended role	Actual contribution to final project
Sagar Tiwari	Lead + full-stack	~99–100% of the working project (see below)
Ambikeshwar	Frontend (early)	Very basic HTML-only page(s); not usable as product UI
Aadrika	Frontend	Partial frontend draft; improvement requested but not delivered

10.2 Sagar Tiwari — primary author (≈ 99–100%)

Almost the **entire final system** was designed, implemented, integrated, tested, and documented by Sagar, including:

- Backend architecture (Express, MongoDB models, APIs, auth, roles, middleware)
- Complete working frontend in `preview/` (React + Vite + Tailwind + dashboards)
- Marketplace, admin / super-admin / support / developer / client panels
- Payments, invoices, wallet, payouts, coupons, settings
- Contact inbox, live chat (guest capture + staff inbox + auto-reply rules)
- Follow-ups / reminders CRM notes
- Reports, charts, Excel & PDF exports
- Seed data, credentials, walkthrough docs, and demo readiness for college

Without this work, the project would not run as a coherent full-stack product.

10.3 Ambikeshwar — early HTML attempt (not used)

- Shared a **very basic HTML-based frontend**
- Quality was **below even a basic usable UI** (static / incomplete; not aligned with the real product)
- That code was **not suitable** for the marketplace, role dashboards, or API integration
- **Not part of the final delivered application** (`preview/` was built instead)

College note: Contribution was limited to an early, inadequate HTML sample; it did not become the project frontend.

10.4 Aadrika — frontend draft (incomplete / not improved)

- Provided a frontend that was **somewhat better** than the basic HTML attempt
- Was clearly asked to **improve and complete** the frontend to product / demo standard

- **Did not deliver the requested improvements**
- Final UI used in the project is the independently completed React app under **preview/** , not that unfinished draft

College note: Partial early frontend help was acknowledged, but ownership of the finished UI and integration remains with Sagar.

10.5 Summary table (for evaluators)

Area	Who delivered the final working version
Project idea / planning docs	Sagar (primary)
Backend API + database	Sagar
Final React UI (preview/)	Sagar
Role dashboards & ops features	Sagar
Payments / invoices / wallet	Sagar
Live chat + contact + follow-ups	Sagar
Reports / Excel / PDF	Sagar
Demo seed + documentation	Sagar
Early basic HTML pages	Ambikeshwar (discarded)
Early incomplete frontend draft	Aadrika (not improved; not final)

Overall estimate: Sagar $\approx$ **99–100%** of the **working college project**. Teammate inputs were early / incomplete frontend attempts that were **not** the basis of the submitted demo.

11. What was intentionally kept simple (college scope)

- Treated as **college / demo project** (not full commercial launch)
- Live Razorpay production, full KYC, legal pages, and production deploy are **out of scope** for this submission (see `docs/LAUNCH_CHECKLIST.md` if asked)
- Ticket-style “Support” module was dropped in favour of **Live Chat** for a clearer demo

12. Learning outcomes

- Full-stack product thinking (roles, marketplace, money flow)
- REST API design with JWT and role middleware
- React SPA structure with protected routes and dashboard UX
- MongoDB modelling for catalogue, CRM, chat, and finance
- Reporting / export (Excel, PDF) and operational UI (inbox, follow-ups, live chat)
- Honest team contribution documentation for academic evaluation

13. Conclusion

SM Global Solution Hub is a working full-stack marketplace demo suitable for college presentation: public catalogue, multi-role panels, sales ops, live chat, and reports.

The **final runnable product** was built almost entirely by **Sagar Tiwari**. Teammates **Ambikeshwar** and **Aadrika** provided early frontend material that was either too basic or left unfinished after improvement was requested; that material is **not** the submitted UI.